

02_PREDICTIVE_DIGITAL_TWIN

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EVIDENCE PROVENANCE: [PREDICTIVE / SIMULATION]

Decision-support lookahead engine evaluated across synthetic and historical NMDC fog scenarios.

1. Decision-Support Role (NOT Central Controller)

The Predictive Digital Twin evaluates what-if scenarios over a 30s to 60s receding horizon. It is NOT an autonomous controller; it generates advisory pacing recommendations for mine operations supervisors to prevent switchback queues and haul road gridlock.

2. What-If Fog Progression Capabilities

- **Fog Front Lookahead:** Models fog rolling into steep switchbacks, reducing optical visibility from 1000m to 50m.
- **Dynamic Headway Pacing:** Computes expanding safe stopping distance and required vehicle separation.
- **Bottleneck Forecast:** Identifies imminent queues at crusher dumping bays and one-lane switchbacks before they form.

3. Separation of Concerns

Capability	Live Digital Twin	Predictive Digital Twin
Purpose	Current reality mirroring	What-if lookahead & queue forecast
Time Domain	Current instant (T = 0)	Future horizon (T+30s to T+60s)
Authority	Authoritative state store	Advisory decision support
Data Source	Physical sensors / live telemetry	Kinematic projection & what-if models